TECHNIQUE MASTERING QUESTIONS

C - FUNCTIONS

1.

a. $D_f = (1,\ \infty)$ $f(5) = \frac{1}{2}$ **b.** $D_f = \mathbb{R}\backslash\{4\}$ $f(1) = 1$

c. $D_f = \mathbb{R}$ $f(3) = 109$ **d.** $D_f = \mathbb{R}$ $f(0) = 1$

e. $D_f = \{x \mid x \neq k\pi,\ k \in \mathbb{Z}\}$ $f(\frac{5\pi}{2}) = 1$

f. $D_f = \mathbb{R}\backslash\{0\}$ $f(\ln 3) = e^{\frac{1}{\ln 3}}$ **g.** $D_f = [0,\infty)\backslash\{1\}$ $f(2) = \frac{\sqrt{2}+1}{\sqrt{2}-1}$

h. $D_f = \{x \mid x \neq (2k+1)\frac{\pi}{2},\ k \in \mathbb{Z}\}$ $f(-\frac{\pi}{4}) = -1$

i. $D_f = \mathbb{R}\backslash\{0\}$ $f(-1) = -1$ **j.** $D_f = \mathbb{R}\backslash\{-1,\ 1\}$ $f(0) = -1$

k. $D_f = (0,\ \infty)\backslash\{1\}$ $f(e) = \frac{1}{e-1}$ **l.** $D_f = \mathbb{R}$ $f(0) = \ln 2$

2.

a. $(f+g)(x) = 2x+5+x^2, x \in \mathbb{R},\ (f-g)(x) = 2x+5-x^2, x \in \mathbb{R},$

$(fg)(x) = (2x+5)x^2, x \in \mathbb{R},\ (\frac{f}{g})(x) = \frac{2x+5}{x^2},\ x \neq 0,\ \ (f \circ g)(x) = 2x^2+5,\ x \in \mathbb{R},$

$(g \circ f)(x) = (2x+5)^2 x \in \mathbb{R},.$

b. $(f+g)(x) = x^2 + \frac{1}{x-1}, x \neq 1,\ \ (f-g)(x) = x^2 - \frac{1}{x-1},\ \ x \neq 1,$

$(fg)(x) = \frac{x^2}{x-1},\ \ x \neq 1,\ \ (\frac{f}{g})(x) = x^2(x-1),\ \ x \neq 1,$

$(f \circ g)(x) = \frac{1}{(x-1)^2},\ \ x \neq 1,\ \ (g \circ f)(x) = \frac{1}{x^2-1},\ \ x \neq -1$ and $x \neq 1.$

c. $(f+g)(x) = e^x + \ln x,\ x > 0,\ \ (f-g)(x) = e^x - \ln x,\ x > 0,\ (fg)(x) = e^x \ln x,\ x > 0,$

$(\frac{f}{g})(x) = \frac{e^x}{\ln x},\ ,\ x > 0$ and $x \neq 1,\ \ (f \circ g)(x) = x,\ \ x > 0,\ \ (g \circ f)(x) = x,\ x \in \mathbb{R}$

d. $(f+g)(x) = \cos x + \frac{1}{x},\ x \neq 0,\ \ (f-g)(x) = \cos x - \frac{1}{x},\ x \neq 0.,$

$(fg)(x) = \frac{\cos x}{x},\ x \neq 0\ ,\ \ (\frac{f}{g})(x) = x\cos x,\ x \neq 0,\ \ (f \circ g)(x) = \cos(\frac{1}{x}),\ x \neq 0,$

$(g \circ f)(x) = \frac{1}{\cos x},\ \ x \neq (2k+1)\frac{\pi}{2},\ k \in \mathbb{Z}.$

e. $(f+g)(x) = \sqrt{x} + x - 1,\ x \geq 0,\ \ (f-g)(x) = \sqrt{x} - x + 1,\ x \geq 0,$

$(fg)(x) = \sqrt{x^3} - \sqrt{x},\ x \geq 0,\ \ (\frac{f}{g})(x) = \frac{\sqrt{x}}{x-1},\ x \geq 0$ and $x \neq 1,$

$(f \circ g)(x) = \sqrt{x-1},\ x \geq 1,\ \ (g \circ f)(x) = \sqrt{x} - 1,\ x \geq 0.$

f. $(f+g)(x) = \sqrt{1-x} + 2^x,\ x \leq 1,\ \ (f-g)(x) = \sqrt{1-x} - 2^x,\ x \leq 1,$

$(fg)(x) = \sqrt{1-x} \times 2^x,\ x \leq 1,\ \ (\frac{f}{g})(x) = \frac{\sqrt{1-x}}{2^x},\ x \leq 1,$

$(f \circ g)(x) = \sqrt{1-2^x},\ x > 0,\ \ (g \circ f)(x) = 2^{\sqrt{1-x}},\ x \leq 1.$